

PH 101
Tutorial 4
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1. Find the equation of the path joining the origin O (0,0) to the point P (1,1) in the xy plane that makes the integral $\int_0^P (y'^2 + yy' + y^2) dx$ stationary.
2. Find the **extremal** of the functional (**Note:** Any solution of the Euler–Lagrange equation is called an extremal of the functional $S[x]$)

(a) $S[x] = \int_1^2 \frac{\dot{x}^2}{4t} dt$ that satisfies the end conditions $x(1) = 5, x(2) = 11$.
(Here $\dot{x} = \frac{dx}{dt}$)

(b) $S[x] = \int_0^\pi (2x \sin t - \dot{x}^2) dt$ that satisfies the end conditions
 $x(0) = 0, x(\pi) = 0$.

3. Using the Euler-Lagrange equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = 0$$

Show that

$$\frac{\partial f}{\partial x} - \frac{d}{dx} \left(f - y' \frac{\partial f}{\partial y'} \right) = 0 \quad (1)$$

[Note that Eq. (1) is known as the ‘**second form**’ of Euler equation]

4. Consider the propagation of light rays in an axially symmetric medium, where, in a system of cylindrical polar coordinates (r, θ, z) the refractive index $n = n(r)$ and the rays lie in the plane $z = 0$. Show that Fermat’s time functional has the form

$$T[r] = \frac{1}{c} \int_{\theta_0}^{\theta_1} n(r^2 + \dot{r}^2)^{\frac{1}{2}} d\theta,$$

where $r = r(\theta)$ is the equation of the path, and $\dot{r}$ means $\frac{dr}{d\theta}$.

Show that the extremals of T satisfy the ODE

$$\frac{n r^2}{(r^2 + \dot{r}^2)^{\frac{1}{2}}} = \text{constant}$$

Show further that, if we write $\dot{r} = r \tan\psi$ where ψ is the angle between the tangent to the ray and the local cylindrical surface $r = \text{constant}$, this equation becomes

$$r n \cos\psi = \text{constant}$$

which is the form of Snell's law for this case.

5. A certain oscillator with generalised coordinate q has Lagrangian

$$L = \dot{q}^2 - 4q^2$$

Verify that $q = \sin 2t$ is a motion of the oscillator, and show from first principle that it makes the action functional $S[q]$ stationary in any time interval $[0, \tau]$.